NIRAL THIRUVIZHA 3.0 · REGIONAL REVIEW

Niranthara

NIRANTHARA — “continuous” in Sanskrit

Know if the treatment is actually working.

Team Lighty
Karthika Krishna M · Catherin Lighty S

Anna University Regional Campus
Tirunelveli

NMNT ID NMNTSTD95000106 · Region: Tirunelveli · Panel 3 · Session AN · Review 07/08/2026

THE PROBLEM

ANANYA IS 24, SIX MONTHS INTO TREATMENT, AND HER PHQ-9 SAYS SHE IS IMPROVING.

She sleeps three hours a night and has not told anyone.
Her next appointment is in five weeks.

50–70% of patients never reach remission.

20–60% drop out — silently, between sessions.

OUR QUESTION *How might we utilize AI chatbots and machine learning to address the challenges of incomplete alleviation of depression symptoms, attrition, and loss of follow-up in mental health treatment?*

STAR*D · Rush et al., *American Journal of Psychiatry*, 2006

WHY IT PERSISTS

Episodic Care, Continuous Illness

One Visit, One Score

A rating scale reports a single total. A patient can improve on the total while the one symptom that matters most — sleep, say — never moves.

Monthly Cadence

Visits are scheduled by the calendar, not by how the patient is actually doing. Whatever happens between them is reconstructed from memory, if it's reported at all.

Silent Accumulation

Relapse risk doesn't wait for the next appointment. It builds in the gap, unwatched, until the calendar happens to catch it — or doesn't.

THE SOLUTION

Continuity Between The Appointments

Niranthara watches the days between appointments — check-ins, journals, sleep and heart-rate from a smartwatch — and tells the clinician who is slipping, before the next visit.

Every other system tells you the patient is getting worse.
Niranthara tells you whether what you did about it helped.



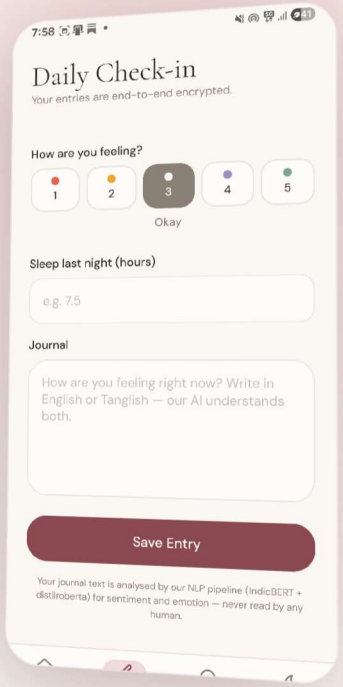
Chat support



Cycle tracking



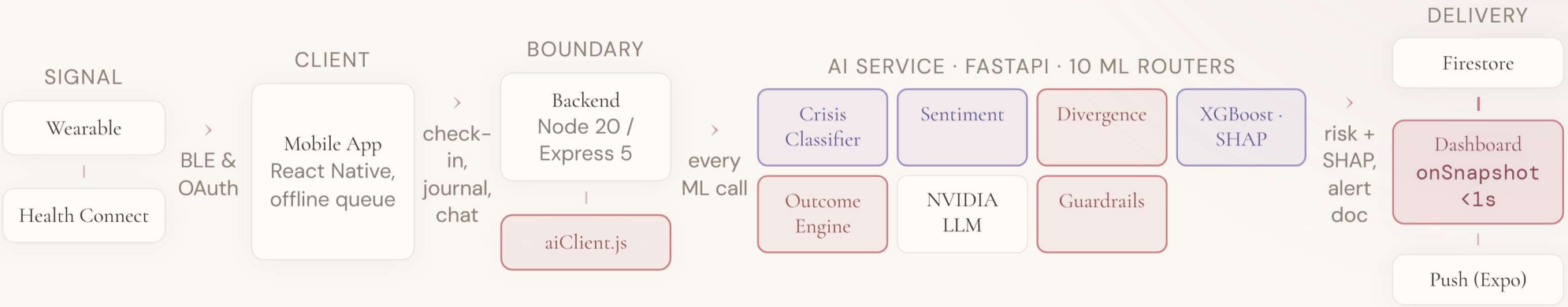
Guided CBT



Live prototype — daily check-in

ARCHITECTURE

Five Services, One Boundary



push notification → patient app · a JITAI nudge closes the loop

THE INTELLIGENCE

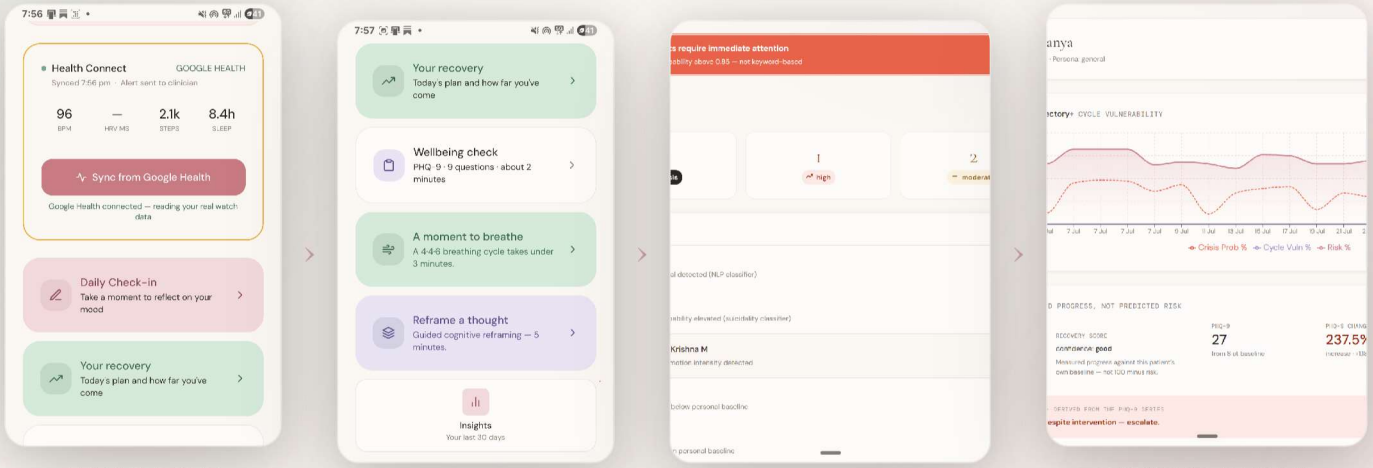
Ten Models, And What Each One Is For

MODEL	TYPE	ROLE & EVIDENCE
Crisis Classifier AI	sentinet/suicidality	Runs locally — works offline. Benign 0.0009 · explicit ideation 0.9945
Sentiment AI	XLM-RoBERTa	Multilingual, returns P(negative). Positive 0.021 · negative 0.929
Mood–Sentiment Divergence Custom	Our algorithm	Self-reported mood vs. language polarity, one scale. Suppression 0.929 · consistency 0.071 — the unique IP
XGBoost Risk + SHAP AI	Gradient boosting	15 features in — risk probability + ranked contributing factors out
Outcome Engine Custom	choose_intervention()	Shrunk mean (n·user+3·pop)/(n+3), bounded exploration 1/√(n+1), two hard safety floors
NVIDIA Cloud LLM Chain	Generation	Llama 3.1 8B (~1.3s) → Nemotron-3 120B (~4.5s) → static fallbacks
Deterministic Guardrails Custom	Input + output checks	Input-side dosing check before the LLM, output-side check after

These seven carry the pipeline end to end, inside a FastAPI service running 10 ML routers in total.

LIVE DEMO

Wearable To Dashboard, Live



WEARABLE
Google Health Connect data

PATIENT APP
Risk score 75, High

CASELOAD VIEW
Ananya flagged high risk

PATIENT DETAIL
30-day trajectory and SHAP factors

Firestore onSnapshot — under a second, no polling · verifyLoop.js — 44 automated checks passed, 0 failed

COMPETITION

Everyone Detects. Nobody Measures.

TYPICAL APPS

Mood trackers — log a number, show a chart

Chatbots — respond in the moment, then forget

Keyword crisis detection — misses code-mixed language

NIRANTHARA

Mood vs. language divergence, seeing suppression a number alone cannot show

Outcome engine — measures whether the chosen intervention worked

Tanglish crisis routing — 0.008 without it, 0.967 with it

MARKET & MODEL

Who Pays, And Why

Complements Tele-MANAS

Government helplines respond when a patient calls. Niranthara watches continuously, so the patient doesn't have to call in crisis for anyone to notice.

B2B2C Distribution

Clinics and counselling institutions deploy Niranthara to their existing patients — we earn trust through the clinician, not around them.

The Compounding Moat

Every clinician-confirmed outcome labels our dataset further. That longitudinal, outcome-labelled data is the asset competitors cannot shortcut.



One-click clinical report for the chart

CLOSE

Every other system tells you the patient is getting worse.

Niranthara tells you whether what you did about it helped.